

Further analogs of Andrews' spt-function identity

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What we study

Main work

We give new analogues of Andrews' celebrated spt-function identity. Instead of using Watson's ${}_8\phi_7$ transformation or Bailey-type ${}_{10}\phi_9$ transformations, we use two ${}_{12}\phi_{11}$ transformations of Verma and Jain between basic hypergeometric series on different bases.

Form of the results

Each new identity has the same structure:

weighted smallest parts-type series = Lambert series term + oscillating term.

Takeaway

The work expands the catalogue of spt-type identities and suggests possible links with mock theta functions and other restricted partitions.

Contents

- 1 Review the smallest parts function $\text{spt}(n)$ and Andrews' original identity.
- 2 Recall previous analogues connected with $\omega(q)$, $\nu(q)$, and $f(q)$.
- 3 State the two new Verma-Jain transformations used.
- 4 State the two new identities.
- 5 Explain the proof mechanism - specialization, limit, rearrangement, and second derivative at $z = 1$.
- 6 Discuss the concluding observations - mock theta questions, further possible analogues, and structural symmetry.

Partitions and the spt-function

A partition $\lambda \vdash n$ is a nonincreasing sequence of positive integers whose sum is n . Andrews' smallest-parts statistic introduced in 2008 counts the total multiplicity of the smallest part across all partitions.

$$\text{spt}(n) = \sum_{\lambda \vdash n} \#\{\text{parts of } \lambda \text{ equal to the smallest part of } \lambda\}.$$

Example ($n = 4$)

The partitions of 4 are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

The smallest-part multiplicities are 1, 1, 2, 2, 4. Hence $\text{spt}(4) = 10$.

It has striking Ramanujan-type congruences -

$$\text{spt}(5n + 4) \equiv 0 \pmod{5},$$

$$\text{spt}(7n + 5) \equiv 0 \pmod{7}, \quad \text{spt}(13n + 6) \equiv 0 \pmod{13}.$$

Notation

q -series (Eulerian series, basic hypergeometric series, q -hypergeometric series) are constructed from the q -rising factorials (q -Pochhammer symbols).

The usual notation for the conventional q -Pochhammer symbols are

$$(a)_n = (a; q)_n := \prod_{k=0}^{n-1} (1 - aq^k), \quad (a)_\infty = (a; q)_\infty := \lim_{n \rightarrow \infty} (a; q)_n,$$
$$(a_1, a_2, \dots, a_j; q)_n = (a_1; q)_n (a_2; q)_n \dots (a_j; q)_n,$$
$$(a_1, a_2, \dots, a_j; q)_\infty = (a_1; q)_\infty (a_2; q)_\infty \dots (a_j; q)_\infty.$$

Generically, q -series take the form ($|q| < 1$)

$$\sum_{n \geq 0} \frac{(a_1; q)_n (a_2; q)_n \dots (a_r; q)_n z^n}{(q; q)_n (b_1; q)_n \dots (b_s; q)_n}.$$

$p(n)$ - number of integer partitions of n .

Generating function for $\text{spt}(n)$

A standard combinatorial argument gives

$$\sum_{n=1}^{\infty} \text{spt}(n)q^n = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2(q^{n+1};q)_{\infty}}.$$

- Choose the smallest part to be n .
- The number of smallest parts contributes $q^n + 2q^{2n} + 3q^{3n} + \cdots = \frac{q^n}{(1-q^n)^2}$.
- Parts larger than n are unrestricted and contribute $(q^{n+1};q)_{\infty}^{-1}$.

Andrews' original spt identity (Crelle journal, 2008)

Andrews' identity is

$$\sum_{n=1}^{\infty} \text{spt}(n) q^n = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^n)^2} q^{n(3n+1)/2}.$$

It is obtained using Watson's q -analogue of Whipple's theorem, a ${}_8\phi_7$ transformation.

$$\begin{aligned} & {}_8\phi_7 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, c, d, e, q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/b, aq/c, aq/d, aq/e, aq^{N+1} \end{matrix}; q, \frac{a^2 q^{N+1}}{bcde} \right) \\ &= \frac{(aq, aq/de; q)_N}{(aq/d, aq/e; q)_N} {}_4\phi_3 \left(\begin{matrix} aq/bc, d, e, q^{-N} \\ aq/b, aq/c, deq^{-N}/a \end{matrix}; q, q \right). \end{aligned}$$

Rank moment interpretation in Andrews' identity

The first term on the right is the generating function for $np(n)$, where $p(n)$ is the partition function -

$$\frac{1}{(q; q)_{\infty}} \sum_{n \geq 1} \frac{nq^n}{1 - q^n}.$$

The second term is the generating function for $\frac{1}{2}N_2(n)$, where

$$N_2(n) = \sum_{m=-\infty}^{\infty} m^2 N(m, n).$$

Here $N(m, n)$ counts partitions of n with rank m .

Rank - statistic due to Freeman Dyson (Eureka, 1944)

Rank of a partition is the largest part minus the number of parts. As in, for $5 + 3 + 3 + 1$, the largest part is 5, the number of parts is 4, and the rank is $5 - 4 = 1$.

Andrews' original spt identity

Andrews' identity is

$$\sum_{n=1}^{\infty} \text{spt}(n) q^n = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^n)^2} q^{3n(n+1)/2}.$$

It is obtained using Watson's q -analogue of Whipple's theorem, an ${}_8\phi_7$ transformation.

$$\begin{aligned} & {}_8\phi_7 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, c, d, e, q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/b, aq/c, aq/d, aq/e, aq^{N+1}; q, \frac{a^2 q^{N+1}}{bcde} \end{matrix} \right) \\ &= \frac{(aq, aq/de; q)_N}{(aq/d, aq/e; q)_N} {}_4\phi_3 \left(\begin{matrix} aq/bc, d, e, q^{-N} \\ aq/b, aq/c, deq^{-N}/a; q, q \end{matrix} \right). \end{aligned}$$

Analogue due to Andrews-Dixit-Yee (Res. Number Theory, 2015)

Andrews, Dixit, and Yee proved an analogue related to $q\omega(q)$ and $\nu(-q)$

$$\sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2 (q^{n+1}; q)_n (q^{2n+2}; q^2)_{\infty}} = \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{2n})}{(1-q^{2n})^2} q^{n(3n+1)}.$$

Context

This came from a limiting case of a $_{10}\phi_9$ transformation due to Bailey, in a form used by Andrews.

$$\begin{aligned} \lim_{N \rightarrow \infty} {}_{10}\phi_9 \left(\begin{matrix} a, q^2 \sqrt{a}, -q^2 \sqrt{a}, p_1, p_1 q, p_2, p_2 q, f, q^{-2N}, q^{-2N+1} \\ \sqrt{a}, -\sqrt{a}, \frac{aq^2}{p_1}, \frac{aq}{p_1}, \frac{aq^2}{p_2}, \frac{aq}{p_2}, \frac{aq^2}{f}, aq^{2N+2}, aq^{2N+1}; q^2, \frac{a^3 q^{4N+3}}{p_1^2 p_2^2 f} \end{matrix} \right) \\ = \frac{(aq; q)_{\infty} \left(\frac{aq}{p_1 p_2}; q \right)_{\infty}}{\left(\frac{aq}{p_1}; q \right)_{\infty} \left(\frac{aq}{p_2}; q \right)_{\infty}} \sum_{n=0}^{\infty} \frac{(p_1; q)_n (p_2; q)_n \left(\frac{aq}{f}; q^2 \right)_n}{(q; q)_n (aq; q^2)_n \left(\frac{aq}{f}; q \right)_n} \left(\frac{aq}{p_1 p_2} \right)^n. \end{aligned}$$

The statistic $\text{spt}_\omega(n)$

This analogue has a partition-theoretic statistic behind it.

$$\sum_{n=1}^{\infty} \text{spt}_\omega(n) q^n = \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)^2 (q^{n+1}; q)_n (q^{2n+2}; q^2)_\infty}$$

The function $\text{spt}_\omega(n)$ is the total number of smallest parts in all partitions of n in which each odd part is less than twice the smallest part. It satisfies the following congruences.

$$\text{spt}_\omega(5n + 3) \equiv 0 \pmod{5},$$

$$\text{spt}_\omega(10n + 7) \equiv 0 \pmod{5},$$

$$\text{spt}_\omega(10n + 9) \equiv 0 \pmod{5}.$$

Mock theta functions in the work of Andrews-Dixit-Yee

The third-order mock theta functions discussed in the paper include

$$\omega(q) = \sum_{n=0}^{\infty} \frac{q^{2n(n+1)}}{(q; q^2)_{n+1}}, \quad \nu(q) = \sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(-q; q^2)_{n+1}}.$$

$$q\omega(q) = \sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)(q^{n+1}; q)_n (q^{2n+2}; q^2)_{\infty}}$$

Recall the other third-order mock theta function

$$f(q) = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(-q; q)_n^2}.$$

Recent work of Andrews and Kumar uses another analogue to connect $f(q)$ with positive odd rank and two-color partitions.

Motivation

New spt identities may reveal new partition statistics or congruences.

Their new analogue is obtained using the limiting case of another $_{10}\phi_9$ -transformation due to Bailey (limiting case due to Andrews).

$$\begin{aligned} & \lim_{N \rightarrow \infty} {}_{10}\phi_9 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, r_1, -r_1, r_2, -r_2, q^{-N}, -q^{-N} \\ \sqrt{a}, -\sqrt{a}, \frac{aq}{b}, \frac{aq}{r_1}, -\frac{aq}{r_1}, \frac{aq}{r_2}, -\frac{aq}{r_2}, aq^{N+1}, -aq^{N+1}; q, -\frac{a^3q^{3+2N}}{br_1^2r_2^2} \end{matrix} \right) \\ &= \frac{(a^2q^2; q^2)_\infty \left(\frac{a^2q^2}{r_1^2r_2^2}; q^2 \right)_\infty}{\left(\frac{a^2q^2}{r_1^2}; q^2 \right)_\infty \left(\frac{a^2q^2}{r_2^2}; q^2 \right)_\infty} \sum_{n=0}^{\infty} \frac{(r_1^2; q^2)_n (r_2^2; q^2)_n (-aq/b; q)_{2n}}{(q^2; q^2)_n (a^2q^2/b^2; q^2)_n (-aq; q)_{2n}} \left(\frac{a^2q^2}{r_1^2r_2^2} \right)^n. \end{aligned}$$

Using this, Andrews and Kumar proved

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{q^{2n}}{(1 - q^{2n})^2 (-q^{n+1}; q)_n (q^{n+1}; q)_\infty} \\ &= \frac{1}{(q; q)_\infty} \sum_{n=1}^{\infty} \frac{nq^{2n}}{1 - q^{2n}} + \frac{1}{(q; q)_\infty} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{2n})^2} q^{3n(n+1)/2}. \end{aligned}$$

Analogue from Andrews-Kumar and implications (PAMS, 2025)

Theorem

Let $N_e(n)$ denote the number of partitions of n with even rank. Then,

$$\sum_{n=1}^{\infty} N_e(n) q^n = \frac{1}{(q; q)_{\infty}} - 2 \sum_{n=1}^{\infty} \frac{q^{2n}}{(1 - q^{2n})(-q^{n+1}; q)_n (q^{n+1}; q)_{\infty}}$$

$$f_3(q) = \frac{1}{(q; q)_{\infty}} - 4 \sum_{n=1}^{\infty} \frac{q^{2n}}{(q^{2n}; q^2)_n (-q^{2n+1}; q)_{\infty}}$$

The partition theoretic interpretation of this is the following. Let $N_o^+(n)$ denote the number of partitions of n with positive odd rank (the rank of a partition is the largest part minus the number of parts), and let $G(n)$ denote the number of partitions into red and blue parts, where the smallest part is even (say $2m$) and all red parts are even and lie within the interval $[2m, 4m]$.

Theorem

$$N_o^+(n) = G(n)$$

The new source - Verma-Jain transformations (JMAA, 1980)

Pattern

The known analogues had been produced using $_{10}\phi_9$ -transformations. We move on to higher-parameter transformations.

We derive new analogues using two $_{12}\phi_{11}$ -transformations of Verma and Jain.

- The first transformation uses parameters grouped as

$$x, \omega x, \omega^2 x, \quad y, \omega y, \omega^2 y,$$

where ω is a primitive cube root of unity.

- The second transformation uses shifted triples

$$x, xq, xq^2, \quad y, yq, yq^2.$$

Why triples matter

Products over the three cube roots of unity factors collapse to base q^3 products, which is why the final identities contain $(\cdot; q^3)_n$ and $(\cdot; q^3)_\infty$.

First Verma-Jain transformation $\Rightarrow$ First spt-function identity analogue

With $\omega^3 = 1$, the first identity we use is

$$\begin{aligned} & {}_{12}\phi_{11} \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, x, \omega x, \omega^2 x, y, \omega y, \omega^2 y, q^{-N}, \omega q^{-N}, \omega^2 q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/x, aq/\omega x, aq/\omega^2 x, aq/y, aq/\omega y, aq/\omega^2 y, aq^{N+1}, \omega^2 aq^{N+1}, \omega aq^{N+1}; q, \frac{a^4 q^{4+3N}}{x^3 y^3} \end{matrix} \right) \\ &= \frac{(a^3 q^3; q^3)_N (a^3 q^3 / x^3 y^3; q^3)_N}{(a^3 q^3 / x^3; q^3)_N (a^3 q^3 / y^3; q^3)_N} {}_6\phi_5 \left(\begin{matrix} q^{-3N}, x^3, y^3, aq, aq^2, aq^3 \\ (aq)^{3/2}, -(aq)^{3/2}, a^{3/2} q^3, -a^{3/2} q^3, x^3 y^3 q^{-3N} / a^3; q^3, q^3 \end{matrix} \right). \end{aligned}$$

The identity (S., Bull. Austral. Math. Soc., 2026)

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1 - q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1 - q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{3n})^2} q^{\frac{n(3n+5)}{2}}$$

- The left side has an spt-type quadratic factor $(1 - q^{3n})^{-2}$.
- The first right-hand series is a Lambert series with $q \mapsto q^3$.

The identity

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1-q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1-q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n)}{(1-q^{3n})^2} q^{\frac{n(3n+5)}{2}}$$

- The summand

$$\frac{q^{3n}}{(1-q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}}$$

has a direct resemblance to Andrews' original summand

$$\frac{q^n}{(1-q^n)^2 (q^{n+1}; q)_{\infty}}.$$

and has a quadratic factor $(1-q^{3n})^{-2}$

- The first right-hand series is a Lambert series with $q \mapsto q^3$.

Second Verma-Jain transformation $\Rightarrow$ Second spt-function identity analogue

The second Verma-Jain identity we use is

$$\begin{aligned}
 & {}_{12}\phi_{11} \left(\begin{matrix} a, q^3 \sqrt{a}, -q^3 \sqrt{a}, x, xq, xq^2, y, yq, yq^2, q^{-N}, q^{-N+1}, q^{-N+2} \\ \sqrt{a}, -\sqrt{a}, aq^3/x, aq^2/x, aq/x, aq^3/y, aq^2/y, aq/y, aq^{N+3}, aq^{N+2}, aq^{N+1} \end{matrix}; q^3, \frac{a^4 q^{3+3N}}{x^3 y^3} \right) \\
 &= \frac{(aq; q)_N (aq/xy; q)_N}{(aq/x; q)_N (aq/y; q)_N} {}_6\phi_5 \left(\begin{matrix} \sqrt[3]{a}, \omega \sqrt[3]{a}, \omega^2 \sqrt[3]{a}, x, y, q^{-n} \\ \sqrt{a}, -\sqrt{a}, \sqrt{aq}, -\sqrt{aq}, xyq^{-n}/a \end{matrix}; q, q \right).
 \end{aligned}$$

Here the transformation goes in the opposite base direction - the ${}_{12}\phi_{11}$ -side is on base q^3 , while the ${}_6\phi_5$ side is on base q .

The identity (S., Bull. Austral. Math. Soc., 2026)

$$\sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)(q^{3n}; q^3)_{\infty} (-q; q)_{n-1} (q; q^2)_n} = \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{3n})}{(1-q^{3n})^2} q^{3n(3n+1)/2}$$

Proof template

Both proofs follow the same template -

- 1 Start from the appropriate Verma-Jain ${}_{12}\phi_{11}$ -transformation.
- 2 Let $a \rightarrow 1$, and set $x = z$, $y = z^{-1}$.
- 3 Use cube-root or shifted-triple products to collapse terms into base q^3 products.
- 4 Let $N \rightarrow \infty$, using an explicit finite-product limit.
- 5 Rearrange the result into a z -dependent identity.
- 6 Apply $-\frac{1}{2} \frac{d^2}{dz^2}$ at $z = 1$.
- 7 Divide by $(q; q)_{\infty}$ or $(q^3; q^3)_{\infty}$ and rewrite the left side.

The derivative identity that creates spt weights

The key calculus identity, due to Andrews, used in both proofs, is

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((1-z)(1-z^{-1})f(z) \right) \Big|_{z=1} = f(1).$$

After the substitutions $x = z$, $y = z^{-1}$, the hypergeometric identities contain products like

$$(z; q^r)_n (z^{-1}; q^r)_n \quad \text{or} \quad (z; q^r)_\infty (z^{-1}; q^r)_\infty.$$

These products have a double zero at $z = 1$. The second derivative extracts the contribution that becomes the square denominator $(1 - q^m)^{-2}$.

Takeaway

This is the mechanism that turns a transformation of basic hypergeometric series into an spt-function identity.

Proof of first identity

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1-q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1-q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n)}{(1-q^{3n})^2} q^{\frac{n(3n+5)}{2}}$$

For $n \geq 1$, the proof uses

$$\lim_{a \rightarrow 1} \frac{(a; q)_n (q\sqrt{a}; q)_n (-q\sqrt{a}; q)_n}{(\sqrt{a}; q)_n (-\sqrt{a}; q)_n} = (q; q)_n (1+q^n).$$

Then in the first Verma-Jain transformation

$$\begin{aligned} & {}_{12}\phi_{11} \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, x, \omega x, \omega^2 x, y, \omega y, \omega^2 y, q^{-N}, \omega q^{-N}, \omega^2 q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/x, aq/\omega x, aq/\omega^2 x, aq/y, aq/\omega y, aq/\omega^2 y, aq^{N+1}, \omega^2 aq^{N+1}, \omega aq^{N+1}; q, \frac{a^4 q^{4+3N}}{x^3 y^3} \end{matrix} \right) \\ &= \frac{(a^3 q^3; q^3)_N (a^3 q^3/x^3 y^3; q^3)_N}{(a^3 q^3/x^3; q^3)_N (a^3 q^3/y^3; q^3)_N} {}_6\phi_5 \left(\begin{matrix} q^{-3N}, x^3, y^3, aq, aq^2, aq^3 \\ (aq)^{3/2}, -(aq)^{3/2}, a^{3/2} q^3, -a^{3/2} q^3, x^3 y^3 q^{-3N}/a^3; q^3, q^3 \end{matrix} \right), \end{aligned}$$

one sets

$$a \rightarrow 1, \quad x = z, \quad y = z^{-1}.$$

This produces factors

$$(z)_n (\omega z)_n (\omega^2 z)_n (z^{-1})_n (\omega z^{-1})_n (\omega^2 z^{-1})_n,$$

which later collapse to base q^3 factors.

Proof of first identity - root-of-unity collapse, the $N \rightarrow \infty$ limit and the key z -identity

The cube-root grouping is essential. Since $\omega^3 = 1$, products over

$$z, \quad \omega z, \quad \omega^2 z$$

collapse into base q^3 expressions. In the proof this is why the final rearranged identity contains

$$(z^3; q^3)_n (z^{-3}; q^3)_n.$$

After replacing z^3 by z , we arrive at expressions in $(z; q^3)_n (z^{-1}; q^3)_n$. The finite truncation terms satisfy

$$\lim_{N \rightarrow \infty} \frac{(q^{-N})_n (\omega q^{-N})_n (\omega^2 q^{-N})_n}{(q^{N+1})_n (\omega^{-1} q^{N+1})_n (\omega^{-2} q^{N+1})_n} q^{3Nn} = (-1)^n q^{3n(n-1)/2}.$$

After the specialization, limit, simplification, and replacement $z^3 \mapsto z$, we obtain

$$\begin{aligned} & \sum_{n=0}^{\infty} \frac{(z; q^3)_n (z^{-1}; q^3)_n (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n} \\ &= \frac{1}{(q^3; q^3)_{\infty}^2} \left((zq^3; q^3)_{\infty} (z^{-1}q^3; q^3)_{\infty} + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n) (z; q^3)_{\infty} (z^{-1}; q^3)_{\infty}}{(1-zq^{3n})(1-z^{-1}q^{3n})} q^{n(3n+5)/2} \right) \end{aligned}$$

Proof of the first identity - derivative evaluations

Key z-identity

$$\sum_{n=0}^{\infty} \frac{(z; q^3)_n (z^{-1}; q^3)_n (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n} \\ = \frac{1}{(q^3; q^3)_{\infty}^2} \left((zq^3; q^3)_{\infty} (z^{-1}q^3; q^3)_{\infty} + \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n) (z; q^3)_{\infty} (z^{-1}; q^3)_{\infty}}{(1-zq^{3n})(1-z^{-1}q^{3n})} q^{n(3n+5)/2} \right)$$

We differentiate the above identity twice using

$$\begin{aligned} -\frac{1}{2} \frac{d^2}{dz^2} \left((z; q^3)_n (z^{-1}; q^3)_n \right) \Big|_{z=1} &= (q^3; q^3)_{n-1}^2, \\ -\frac{1}{2} \frac{d^2}{dz^2} \left(\frac{(z; q^3)_{\infty} (z^{-1}; q^3)_{\infty}}{(1-zq^{3n})(1-z^{-1}q^{3n})} \right) \Big|_{z=1} &= \frac{(q^3; q^3)_{\infty}^2}{(1-q^{3n})^2}, \\ -\frac{1}{2} \frac{d^2}{dz^2} \left((zq^3; q^3)_{\infty} (z^{-1}q^3; q^3)_{\infty} \right) \Big|_{z=1} &= (q^3; q^3)_{\infty}^2 \sum_{n=1}^{\infty} \frac{nq^{3n}}{1-q^{3n}}. \end{aligned}$$

All follow from

$$-\frac{1}{2} \frac{d^2}{dz^2} \left((1-z)(1-z^{-1})f(z) \right) \Big|_{z=1} = f(1).$$

Proof of the first identity

The differentiated identity is

$$\sum_{n=1}^{\infty} \frac{(q^3; q^3)_{n-1}^2 (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n} = \sum_{n=1}^{\infty} \frac{nq^{3n}}{1 - q^{3n}} + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{3n})^2} q^{n(3n+5)/2}.$$

The left side after division by $(q; q)_{\infty}$ is

$$\frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(q^3; q^3)_{n-1}^2 (q; q)_{3n}}{(q^3; q^3)_{2n} (q^3; q^3)_n} q^{3n}.$$

Product cancellation rewrites it as

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1 - q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}}.$$

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1 - q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1 - q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^n)}{(1 - q^{3n})^2} q^{\frac{n(3n+5)}{2}}$$

Proof of the second identity

For the second identity, the proof uses two limits.

$$\lim_{a \rightarrow 1} \frac{(a; q^3)_n (q^3 \sqrt{a}; q^3)_n (-q^3 \sqrt{a}; q^3)_n}{(\sqrt{a}; q^3)_n (-\sqrt{a}; q^3)_n} = (q^3; q^3)_n (1 + q^{3n}).$$

$$\lim_{a \rightarrow 1} \frac{(\sqrt[3]{a}; q)_n (\omega \sqrt[3]{a}; q)_n (\omega^2 \sqrt[3]{a}; q)_n}{(\sqrt{a}; q)_n (-\sqrt{a}; q)_n} = \frac{(q^3; q^3)_{n-1}}{(q^2; q^2)_{n-1}}.$$

The finite truncation terms satisfy

$$\lim_{N \rightarrow \infty} \frac{(q^{-N}; q^3)_n (q^{-N+1}; q^3)_n (q^{-N+2}; q^3)_n}{(q^{N+3}; q^3)_n (q^{N+2}; q^3)_n (q^{N+1}; q^3)_n} q^{3Nn} = (-1)^n q^{3n(3n-1)/2}.$$

After specialization and simplification, we obtain

$$\begin{aligned} & 1 + \sum_{n=1}^{\infty} \frac{(z; q)_n (z^{-1}; q)_n (q^3; q^3)_{n-1}}{(q; q)_n (q^2; q^2)_{n-1} (q; q^2)_n} q^n \\ &= \frac{1}{(q; q)_{\infty}^2} \left((zq; q)_{\infty} (z^{-1}q; q)_{\infty} + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n}) (z; q)_{\infty} (z^{-1}; q)_{\infty}}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} q^{3n(3n+1)/2} \right). \end{aligned}$$

The three derivatives used are

$$\begin{aligned} -\frac{1}{2} \frac{d^2}{dz^2} ((z; q)_n (z^{-1}; q)_n) \Big|_{z=1} &= (q; q)_{n-1}^2, \\ -\frac{1}{2} \frac{d^2}{dz^2} \left(\frac{(z; q)_\infty (z^{-1}; q)_\infty}{(1 - zq^{3n})(1 - z^{-1}q^{3n})} \right) \Big|_{z=1} &= \frac{(q; q)_\infty^2}{(1 - q^{3n})^2}, \\ -\frac{1}{2} \frac{d^2}{dz^2} ((zq; q)_\infty (z^{-1}q; q)_\infty) \Big|_{z=1} &= (q; q)_\infty^2 \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n}. \end{aligned}$$

Differentiating twice gives

$$\sum_{n=1}^{\infty} \frac{(q; q)_{n-1}^2 (q^3; q^3)_{n-1}}{(q; q)_n (q^2; q^2)_{n-1} (q; q^2)_n} q^n = \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} + \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n})}{(1 - q^{3n})^2} q^{3n(3n+1)/2}.$$

Dividing by $(q^3; q^3)_\infty$ and simplifying products gives the identity.

$$\sum_{n=1}^{\infty} \frac{q^n}{(1 - q^n)(q^{3n}; q^3)_\infty (-q; q)_{n-1} (q; q^2)_n} = \frac{1}{(q^3; q^3)_\infty} \sum_{n=1}^{\infty} \frac{nq^n}{1 - q^n} + \frac{1}{(q^3; q^3)_\infty} \sum_{n=1}^{\infty} \frac{(-1)^n (1 + q^{3n})}{(1 - q^{3n})^2} q^{3n(3n+1)/2}$$

The two $_{12}\phi_{11}$ analogues - a comparison with the $_{10}\phi_9$ analogues

$$\sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2 (q^{n+1}; q)_n (q^{2n+2}; q^2)_{\infty}} = \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^2; q^2)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{2n})}{(1-q^{2n})^2} q^{n(3n+1)}$$

$$\sum_{n=1}^{\infty} \frac{q^{2n}}{(1-q^{2n})^2 (-q^{n+1}; q)_n (q^{n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{2n}}{1-q^{2n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n)}{(1-q^{2n})^2} q^{3n(n+1)/2}$$

There is a visible symmetry between the new identities below and the identities above.

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{(1-q^{3n})^2 (q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} = \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^{3n}}{1-q^{3n}} + \frac{1}{(q; q)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^n)}{(1-q^{3n})^2} q^{\frac{n(3n+5)}{2}}$$

$$\sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)(q^{3n}; q^3)_{\infty} (-q; q)_{n-1} (q; q^2)_n} = \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} + \frac{1}{(q^3; q^3)_{\infty}} \sum_{n=1}^{\infty} \frac{(-1)^n (1+q^{3n})}{(1-q^{3n})^2} q^{3n(3n+1)/2}$$

The second sum on the RHS of the identity above resembles the second sum in the first analogue after a formal change of scale.

Concluding observations - possible mock theta or other links

In earlier works, nonweighted companions of spt identities led to mock theta functions.

$$q\omega(q) = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)(q^{n+1}; q)_n (q^{2n+2}; q^2)_{\infty}}$$
$$f_3(q) = \frac{1}{(q; q)_{\infty}} - 4 \sum_{n=1}^{\infty} \frac{q^{2n}}{(q^{2n}; q^2)_n (-q^{2n+1}; q)_{\infty}}$$

Proofs make use of the following four-parameter deep identity due to Andrews.

$$\begin{aligned} & \sum_{n=0}^{\infty} \frac{(B; q)_n (-Abq; q)_n q^n}{(-aq; q)_n (-bq; q)_n} \\ &= \frac{-a^{-1}(B; q)_{\infty} (-Abq; q)_{\infty}}{(-bq; q)_{\infty} (-aq; q)_{\infty}} \sum_{m=0}^{\infty} \frac{(A^{-1}; q)_m \left(\frac{Abq}{a}\right)^m}{\left(-\frac{B}{a}; q\right)_{m+1}} \\ &+ (1+b) \sum_{m=0}^{\infty} \frac{(-a^{-1}; q)_{m+1} \left(-\frac{ABq}{a}; q\right)_m (-b)^m}{\left(-\frac{B}{a}; q\right)_{m+1} \left(\frac{Abq}{a}; q\right)_{m+1}}. \end{aligned}$$

Concluding observations - A nonweighted companion to first identity

Do the new identities have similar mock theta or other partition-theoretic companions?

Main difficulty

The natural nonweighted versions now have more parameters, so Andrews' four-parameter identity does not immediately simplify them.

Removing one factor $(1 - q^{3n})^{-1}$ from the LHS leads to the simplification

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{q^{3n}}{(1 - q^{3n})(q^{3(n+1)}; q^3)_n (q^{3n+1}; q)_{\infty}} &= \sum_{n=0}^{\infty} \frac{q^{3n+3}}{(q^{3n+3}; q^3)_{n+2} (q^{3n+4}; q)_{\infty}} \\ &= \frac{q^3}{(q; q)_{\infty}} \sum_{n=0}^{\infty} \frac{(q; q)_{3n+3}}{(q^{3n+3}; q^3)_{n+2}} q^{3n} \\ &= \frac{q^3}{(q; q)_{\infty}} \sum_{n=0}^{\infty} \frac{(q^3; q^3)_n (q; q^3)_{n+1} (q^2; q^3)_{n+1}}{(q^3; q^6)_{n+1} (-q^3; q^3)_{n+1}} q^{3n}.\end{aligned}$$

Higher-parameter identities analogous to Andrews' four-parameter identity may be needed to deal with this q -series. See G. E. Andrews et al., Overpartitions related to the mock theta function $\omega(q)$, *Acta Arith.* **181** (2017), no. 3, 253–286.

Concluding observations - more analogues to discover

There are several possible sources for further identities -

- additional Verma-Jain identities;
- a ${}_{14}\phi_{13}$ -identity and a related higher-parameter identity;
- identities due to McLaughlin and Zimmer arising from the WP-Bailey tree;
- multisum generalizations of Verma-Jain's ${}_{14}\phi_{13}$ -identities.

Summary of the mathematical narrative

- 1 Andrews' $\text{spt}(n)$ connects smallest-part weights with rank moments and modular phenomena.
- 2 Earlier analogues came from $_{10}\phi_9$ -transformations and led to mock theta and two-colour partition interpretations.
- 3 We obtain two new analogues from Verma-Jain $_{12}\phi_{11}$ -transformations.
- 4 The observations point toward possible mock theta connections, higher-parameter identities, and refined partition interpretations.

Possible discussion questions

- Can the two new left-hand sides be given natural partition-theoretic interpretations?
- Do the nonweighted companion series connect to known or new mock theta functions?
- Is there a general framework explaining the symmetry among Andrews-type spt identities?

Thank you!

References



G. E. Andrews, Ramanujan's "lost" notebook IV. Stacks and alternating parity in partitions, *Adv. Math.* 53 (1984), 55–74.



G. E. Andrews, The number of smallest parts in the partitions of n , *J. Reine Angew. Math.* 624 (2008), 133–142.



G. E. Andrews, A. Dixit and A. J. Yee, *Partitions associated with the Ramanujan/Watson mock theta function $\omega(q)$, $\nu(q)$ and $\phi(q)$* , *Res. Number Theory* 1 (2015), 1–25.



G. E. Andrews, A. Dixit, D. Schultz and A. J. Yee, Overpartitions related to the mock theta function $\omega(q)$, *Acta Arith.* 181 (2017), 253–286.



G. E. Andrews and R. Kumar, Rank, two-color partitions and mock theta functions, *Proc. Amer. Math. Soc.* 153 (2025), 4669–4682.



R. Sarma, Further analogues of Andrews' spt-function identity, *Bull. Aust. Math. Soc.*, published online 2026.



A. Verma and V. K. Jain, Transformations between basic hypergeometric series on different bases and identities of Rogers-Ramanujan type, *J. Math. Anal. Appl.* 76 (1980), 230–269.